Circuit investigations

Construct safely

1 Measurement circuit

Low-voltage supply, switch, ammeter and test component in series. Voltmeter across test component.

Variable resistor controls current. Never put an ammeter directly across a supply.

2 Quality & safety

Record matched V and I; calculate R = V/I. Secure connections; check meter ranges and zero.

Repeat; avoid hot wire; switch off between readings. A repeated zero error is not removed by averaging.

Compare resistance

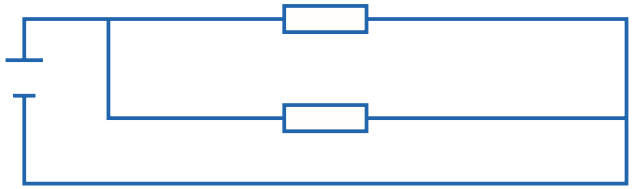
3 Length of wire

Change measured wire length using movable contact and ruler.

Keep material, cross-sectional area and temperature fixed. Low current limits heating.

R vs length should be proportional for uniform wire at constant temperature.

4 Combinations



Parallel paths share the same pd

Measure each resistor, then the same pair in series and parallel.

Series total = sum. Parallel total is less than either. Extra parallel path increases total current at fixed V.

Plot & explain

5 I–V method

Vary pd, record paired I/V readings across a useful range. Reverse supply for negative values.

Repeat for resistor, lamp and diode; protective series resistor for diode. Stay within ratings.

6 Graph conclusions

Ohmic: straight through origin if temperature constant. Lamp: flatter at larger |V| as it heats.

Diode: forward conduction, tiny reverse current. Draw smooth best-fit curves; do not connect noisy points mechanically.